INCH-POUND
MIL-DTL-16377/31B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/31A(SH)
9 May 1983

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT AND SOLID-STATE EQUIVALENT, DETAIL LIGHTING FOR SPOT ILLUMINATION, 6-25 WATTS, 120 VOLTS, TOTALLY ENCLOSED, SYMBOLS 147.2 AND 147.2SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 8 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

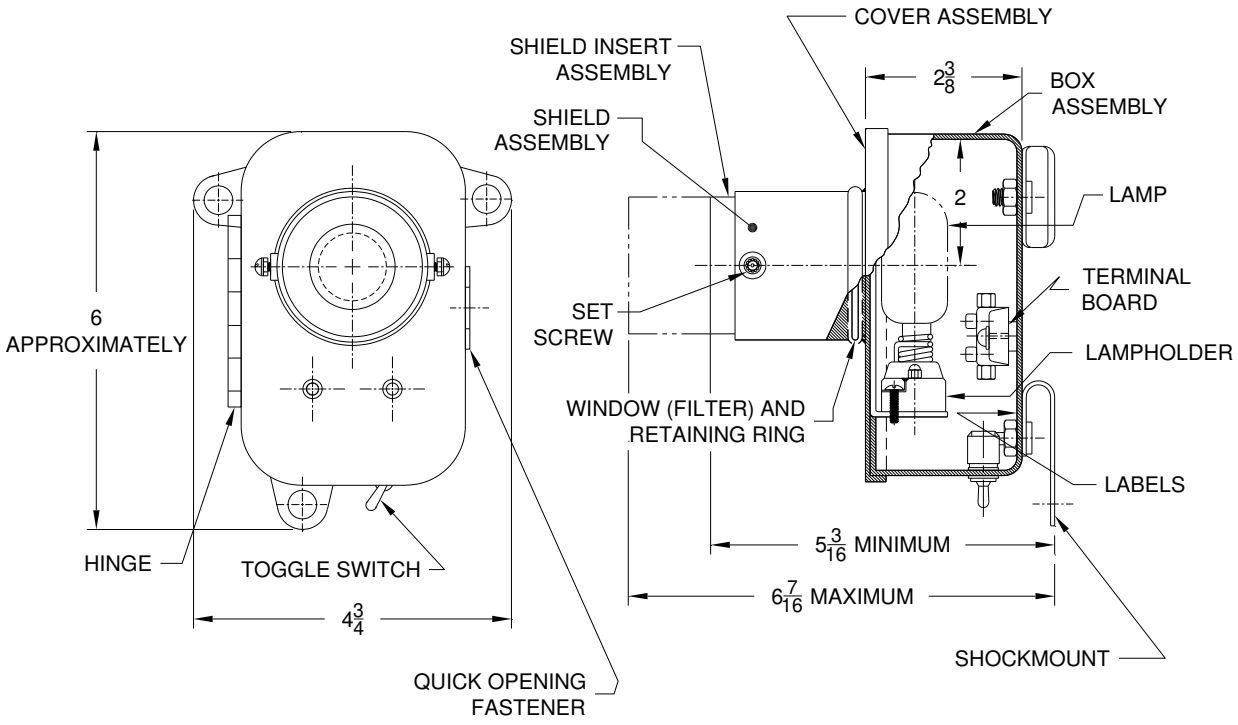


FIGURE 1. Typical view of symbol 147.2.

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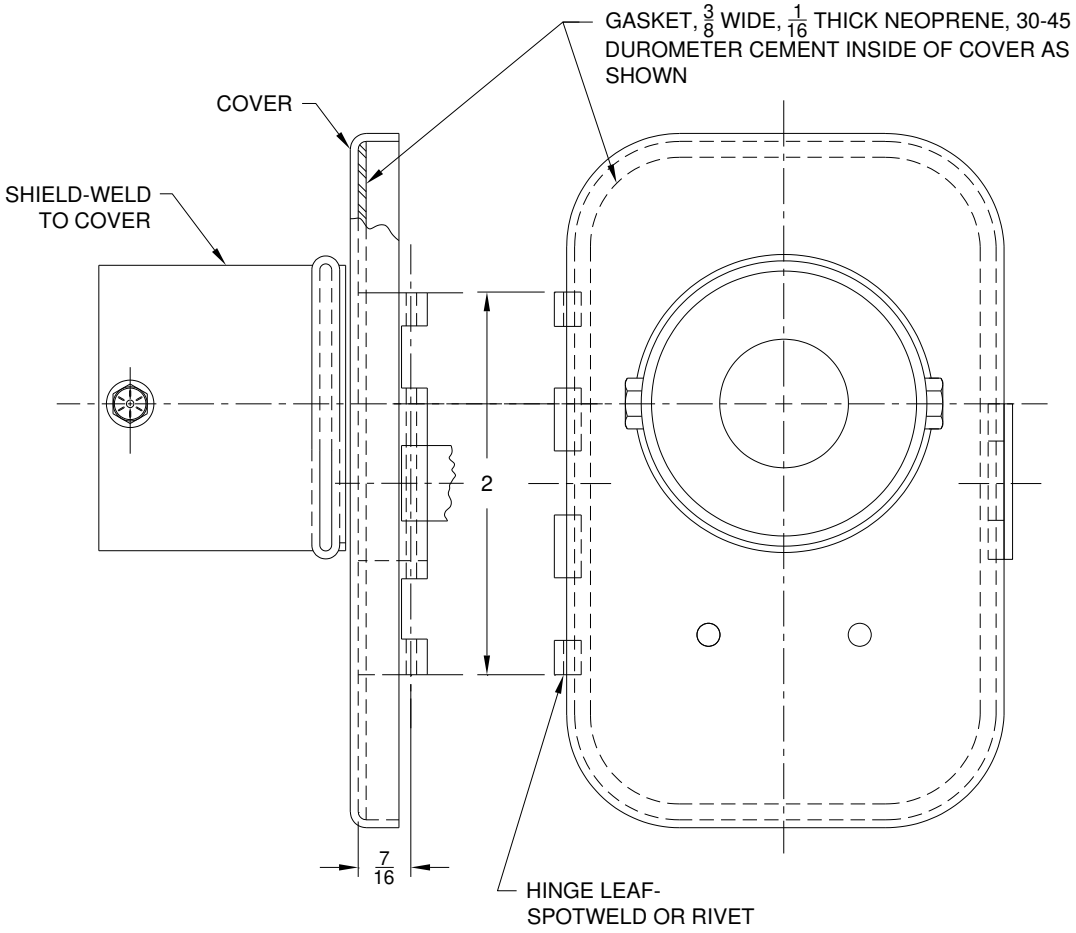


FIGURE 2. Cover assembly.

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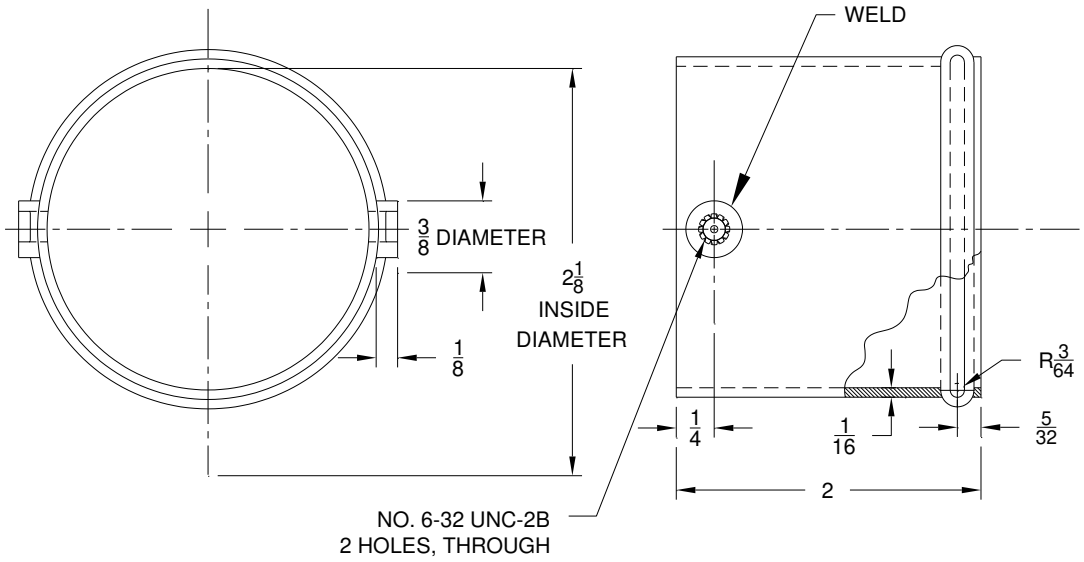


FIGURE 3. Shield assembly.

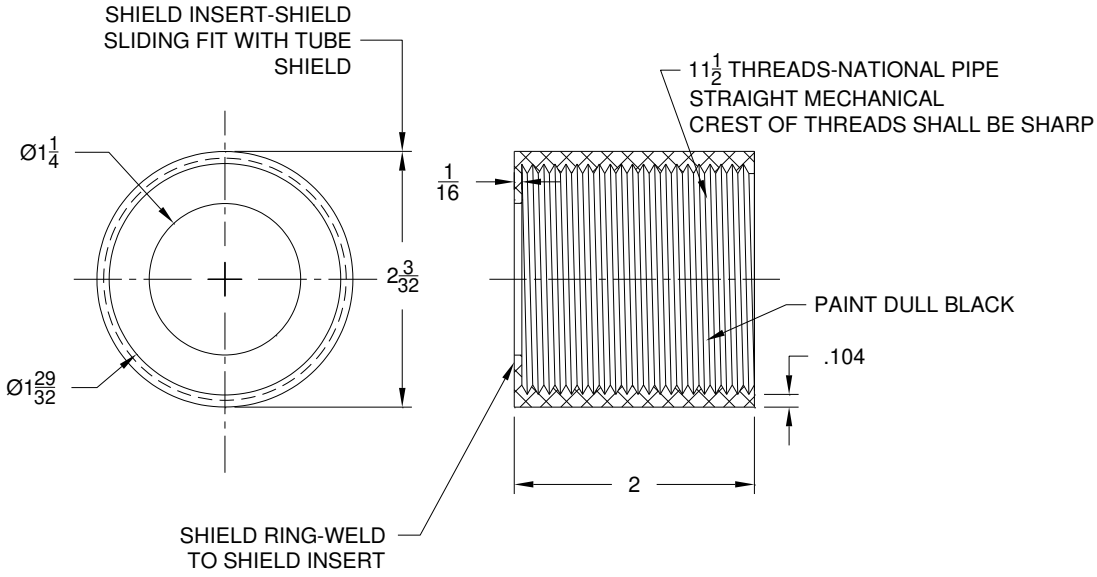


FIGURE 4. Shield insert.

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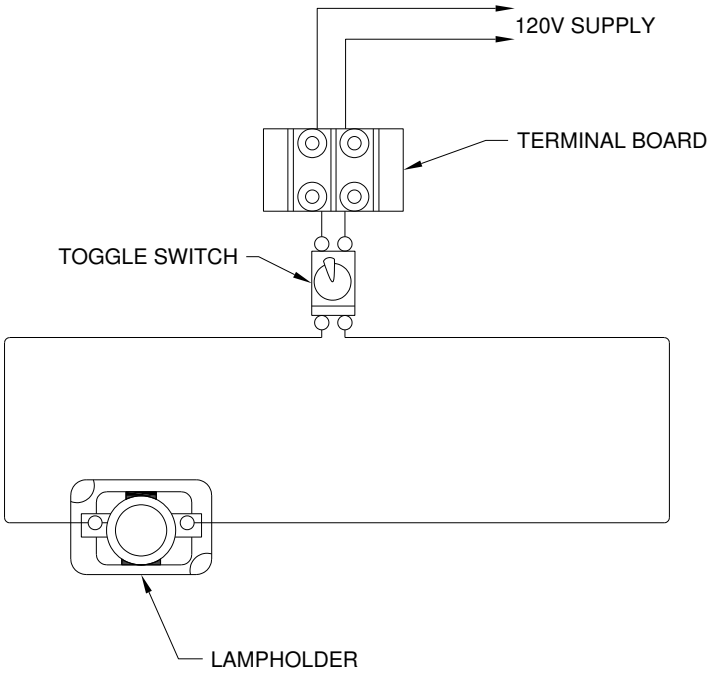


FIGURE 5. Wiring diagram.

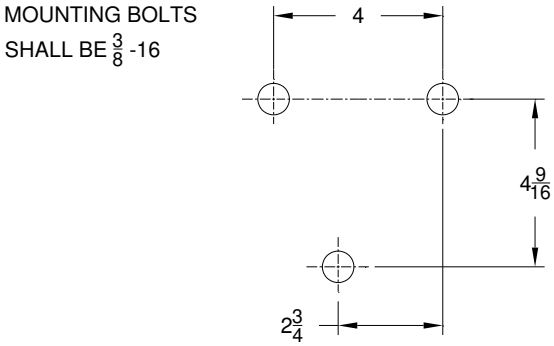


FIGURE 6. Mounting dimensions.

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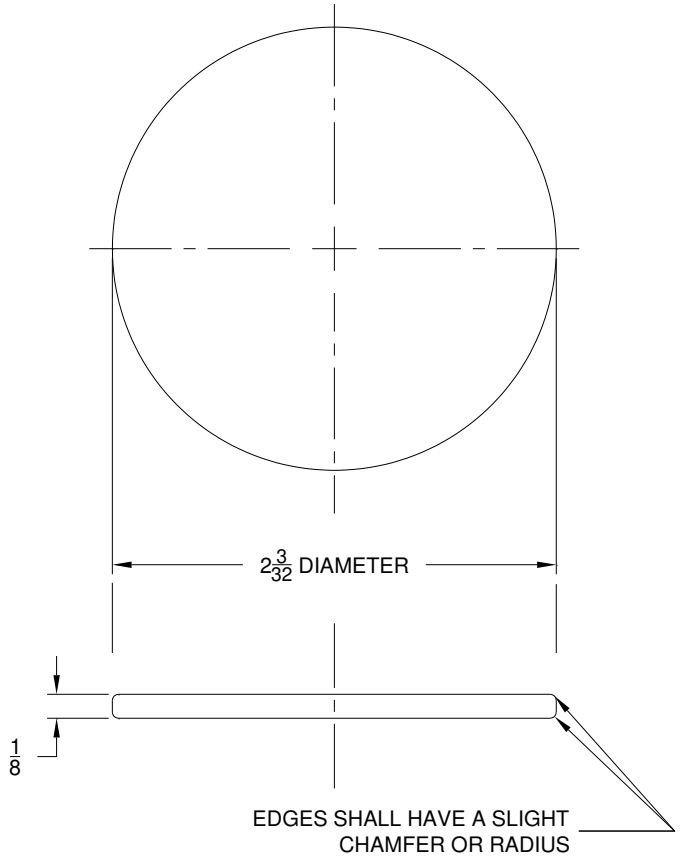


FIGURE 7. Window filter.

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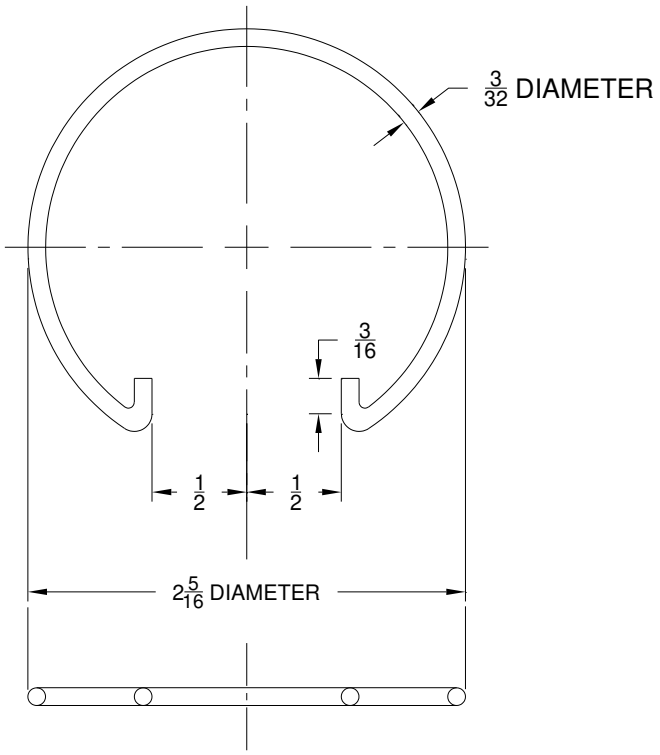


FIGURE 8. Retaining ring.

INCANDESCENT REQUIREMENTS

CLASSIFICATION

Type: II
Class: 1

SYMBOL NUMBER

147.2: Light, detail, for use with 6-watt, 15-watt, or 25-watt clear (white) lamps for use in white, red, and blue illumination.

DESIGN

Electrical rating:	6, 15, or 25 watts, 120 volts.
Dimensions and configuration:	See figures 1 through 4 and 6 through 8.
Enclosure, including cover assembly, box assembly, shield assembly, and shield insert assembly:	Aluminum. Interior surfaces of enclosure shall be baked white enamel. Interior surfaces of shield assembly and shield insert assembly shall be painted dull black.
Shock mounts:	MIL-F-16377/41-010.
Weight:	1 pound, 4 ounces maximum.
Hinge:	Aluminum.
Fastener, quick opening:	Size to suit. Dzus Manufacturing Company Incorporated, or equal.
Lampholder mounting bracket:	Steel, zinc plated.
Lampholder:	UL 496, Supplement SB, candelabra screw.

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Switch:	Toggle DPST, MS16569-1.
Wiring diagram:	See figure 5.
Terminal board:	Commercial type.
Window (filter), red:	See figure 7. Material shall be red transparent plastic sheet, type RTPS-1 of MIL-DTL-24191.
Window (filter), blue:	See figure 7. Material shall be red transparent plastic sheet, type BTPS-1 of MIL-DTL-24191.
Retaining ring:	See figure 8. Material shall be spring steel.
Set screw:	Corrosion-resistant steel.
Lamp:	These fixtures shall be designed to accommodate the following lamps (lamps not furnished): (a) 6-watt conforming to industry number 6S6(120). (b) 15-watt conforming to industry number 15T7C. (c) 25-watt conforming to industry number 25T8C.
Label:	Labels shall be furnished in accordance with MIL-DTL-16377. One type IV label shall be applied as shown on figure 1. The information to be shown on the type IV label shall be: "INSTALL THE FOLLOWING LAMPS ONLY: 6-WATT: NSN 6240-00-143-3049 15-WATT: NSN 6240-00-617-1719 25-WATT: NSN 6240-00-143-3057"

SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 1

SYMBOL NUMBER

147.2SSL: Light, detail, for use in white, red, and blue illumination.

DESIGN

Electrical rating:	13 watts, 120 volts, 60 Hertz.
Maximum dimensions:	See figures 2 and 4.
Wiring and controls:	Shall be capable of providing output equivalent to an incandescent fixture with either a 6-watt, 15-watt, or 25-watt lamp, selectable on the fixture.
Photometric data:	Three selectable states with illumination equivalent to an incandescent 6-watt, 15-watt, or 25-watt lamp.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.
Weight:	1 pound, 4 ounces maximum.
Designation and marking:	One type I label shall be furnished in accordance with MIL-DTL-16377.
Material:	Corrosion-resistant in accordance with MIL-DTL-16377.

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EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:
Complete fixtures: See table I for type II and table II for type III.
Conformance inspection:
Comparison inspection: Same as the first article inspection.
Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION

(a) Assignment of military part numbers and general information is shown in table III.
(b) Retaining ring and red and blue window (filter) shall not be installed in the light. They shall be furnished with the light in a bag.

TABLE I. First article and comparison inspection for complete type II fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14	Totally enclosed
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Continuity of grounding	3.6.13.4	4.8.12	---
Magnetic permeability	3.4.8	4.8.18	---

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TABLE II. First article and comparison inspection for complete type III fixtures, symbol 142.2SSL.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Power interface	3.6.18	4.8.21	---
Shock	3.5.12	4.8.7	Grade B 1/
Vibration	3.5.13	4.8.8	1/
Brightness	3.7.5	4.8.6.4	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Photometrics	3.7.1	4.8.6	See requirements section
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.6.17	4.8.22	---
Stress relief	3.8.5	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---
Continuity of Grounding	3.6.13.4	4.8.12	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

TABLE III. Part number and general information.

Military Part Number M16377/31-	Item Name	National Stock Number (NSN)	Supersedes Part Number
147.2	Complete type II fixture, symbol 147.2	---	MS17316-147.2
001	Red filter	6210-00-284-0480	MS17153-17
002	Blue filter	---	New
147.2SSL	Complete type III fixture	---	---

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-012)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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